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Agile Software Development

Managing User Stories with **Scrum** and **Kanban** Frameworks

🔄 Iterative Focus (Scrum)

🔄 Continuous Flow (Kanban)

PART 1

Scrum Methodology

An iterative and incremental framework based on fixed time periods called Sprints.

Scrum: Core Pillars & Principles



1. Transparency

All process details are visible to everyone responsible for outcomes (Product Owner, Scrum Master, and Developers).



2. Inspection

Regular evaluation of Scrum artifacts and Sprint progress to spot unexpected issues early.



3. Adaptation

Immediate process or feature adjustments if inspection shows unacceptable deviations.

Scrum: How to Build a User Story

Key Elements (3C Model)

Card: Standard format: *"As a [Role], I want [Feature], so that [Benefit]"*.

Conversation: Continuous dialogue between Product Owner and Developers to clarify requirements.

Confirmation: Clear Acceptance Criteria (Given-When-Then format).

Estimation: Story Points assigned using Planning Poker.

Workflow and Inclusion

1. Product Backlog Entry: Product Owner writes the User Story based on user needs.

2. Refinement (Grooming): The team clarifies details and breaks the story into technical tasks.

3. Estimation & Priority: Assigning complexity points and business value priority.

4. Sprint Commitment: Moved into the Sprint Backlog during Sprint Planning.

Scrum: Tools for User Stories

-  **Jira Software:** Industry standard for managing Product Backlogs, Sprints, and Scrum boards.
-  **Azure DevOps:** Connects user stories directly with Git code repositories and CI/CD pipelines.
-  **ClickUp / Trello:** Lightweight tools ideal for smaller teams or academic project setups.
-  **Miro / Mural:** Visual boards used for collaborative story mapping and team refinement.

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Scrum: Tracking a User Story



Daily Scrum

A 15-minute daily meeting. Developers review progress, daily goals, and any current blockers.



Burndown Chart

Graph showing remaining work versus time, indicating if stories will finish on schedule.



Definition of Done (DoD)

Criteria required to consider a story finished: code reviewed, tested, and ready for release.

PART 2

Kanban Methodology

A visual management framework focused on continuous workflow and team capacity optimization.

☐ Kanban: Key Principles & WIP



1. Visualize Workflow

Mapping all development steps (To Do, In Progress, Testing, Done) on a clear visual board.



2. Limit WIP

Work In Progress (WIP): Restricts max active tasks per column to prevent team overload.



3. Manage Flow

Tracking movement speed across stages to find and fix process bottlenecks quickly.

Kanban: How to Build a User Story

Card Details

Short Title: Concise name for the user story requirement.

Class of Service: Work category by urgency (Expedite, Fixed Date, Standard).

Explicit Policies: Clear rules to transition cards between board columns.

Dynamic Assignment: Pulled by team members based on capacity (Pull System).

Just-In-Time Process

- 1. Backlog Arrival:** User Stories enter the queue ordered by business priority.
- 2. Just-In-Time Refinement:** Stories are detailed right before work begins.
- 3. PULL System:** Developers pull new stories only when WIP capacity opens up.

Kanban: Visual Tools

-  **Trello:** Lightweight visual board perfect for simple task columns and tags.
-  **Kanbanize / Businessmap:** Enterprise platform built for advanced flow metrics.
-  **Jira Kanban:** Configured for continuous delivery, WIP limits, and analytics.
-  **LeanKit / Asana:** Helps model multi-team workflows and card dependencies.

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🕒 Kanban: Tracking & Metrics

🕒 Key Time Metrics

Lead Time: Total duration from initial user request until final delivery.

Cycle Time: Duration spent actively developing the story.

$$\text{Lead Time} = \text{Wait Time} + \text{Cycle Time}$$

📈 Flow Monitoring

Cumulative Flow Diagram (CFD): Tracks tasks in each stage over time to highlight bottlenecks.

Blocker Focus: The team prioritizes unblocking cards before starting new tasks.

Kanban Standup: Daily board review conducted from right to left (closest to Done first).

Summary: Scrum vs. Kanban

Criterion	Scrum	Kanban
Cadence / Delivery	Fixed Sprints (1 to 4 weeks).	Continuous delivery flow.
Defined Roles	Product Owner, Scrum Master, Developers.	No mandatory predefined roles.
Key Metrics	Sprint Velocity, Burndown Chart.	Lead Time, Cycle Time, Throughput.
Change Management	No changes allowed during active Sprint.	Changes allowed anytime if WIP permits.
Story Estimation	Mandatory (Story Points / Hours).	Optional (Focuses on uniform card size).

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Questions or Comments?

Thank you for your attention!



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Agile Methodologies